

Indranil Maiti

Single Photon Applications Laboratory, Nicolaus Copernicus University, Torun, Poland

✉ indranilmaiti16@gmail.com

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Education

- 2022 – Ongoing **Ph.D., Nicolaus Copernicus University, Torun, Poland**
Secure Quantum Communication in multiplexed optical network
- 2019 – 2021 **M.Sc. Physics, Visva Bharati University**
Special paper: Quantum optics, Atomic Molecular Physics
- 2016 – 2019 **B.Sc. Physics, Visva-Bharati University**
Elective: Electronics and engineering.

Publications

- 1 I. Maiti and M. Lasota, “Advantages and limitations of channel multiplexing for discrete-variable entanglement-based quantum key distribution,” *New Journal of Physics*, vol. 27, no. 3, 2025.

Internships

- April. 2024 – September. 2024 **Photonics Intern** amPICQ Pvt. Ltd.
Photonics Set-up design for DPS QKD
I did a comprehensive research on analyzing and requirements to design a QKD set-up for a DPS (Differential Phase Shift) Quantum Key Distribution set-up.

Certificate

- May. 2023 – June. 2023 **Intern student** QNu Labs Pvt. Ltd.
Automation of the optical path for Armos QKD
I wrote script (in Python) for automating the instruments used in Armos QKD set-up at QNu for remote use. I won the 2nd Prize and a cash prize of 25k (INR) in SparQ Internship Programme.

Certificate

- Sept. 2022 – Dec 2022 **Intern student** Qulabs Software India Pvt. Ltd.
Quantum error mitigation in measurement-based quantum computation
I compared different error mitigation and error suppression techniques used in IBM simulator with different error models and with different circuit depth. I compared results using zero-noise extrapolation (ZNE) for different quantum algorithms and benchmark the results.

Certificate

Internships (continued)

March. 2022 – June. 2022

Intern student C-DAC India

Specification of single photon source for satellite-based quantum communication

I did a literature review on existing satellite-based quantum communication schemes and calculated the crystal parameters for generating entangled photon pair using Spontaneous Parametric Down Conversion (SPDC) for an experiment.

Certificate

Conferences and Poster Presentation

2024

Central European Workshop on Quantum Optics, Palacky University, Olomouc, Czech Republic

Summer School in Quantum Science and Technologies, University of Bari, Italy

International Conference on Quantum Communication, Measurement and Computing (QCMC) IIT Madras, Chennai,.

2025

Quantum Optics XI, Krakow, Poland.

2026

Single Photon Workshop, Naples, Italy.

Qcrypt, Ottawa, Canada.

Certifications, Awards and Achievements

2023

IBM Quantum Challenge: Spring, 2023.

Certificate

2021

Summer School on quantum information and quantum Technology, 2021. Awarded by IISER Kolkata, India.

Faculty Development Program on quantum computing. Awarded by IIIT Dharwad, India.

Q-Bronze: Quantum Computing and programming workshop. Awarded by QWorld.

You will get all my certificates here

2016

INSPIRE Scholarship. Govt. of India.

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Skills

Coding

C++, Python, Javascript, Mathematica

Software Development

Full Stack Web Developer (MERN), Database Design, UI/UX Design

Languages

Bengali (Native Speaker), Hindi, English

TOEFL Score